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Effects of Material Properties of Cue on Ball Trajectory in Billiards*

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Abstract

The numerical evaluation of impact characteristics of a cue in billiards is performed for the case where the ball trajectory is deviated from the hitting direction and curved by hitting a ball on its upper left or right part with an inclined cue. The effective numerical method developed in the previous papers, in which a cue and a ball are assumed to be isotropically elastic and rigid, respectively, is extended for this case. The extended method is verified with an experiment using a high speed camera. The result obtained with the extended method shows that the radius of curvature of the ball trajectory is large for the impact on its almost exactly left part with a slightly inclined cue, and this result agrees with an empirical one. It is found from a numerical simulation that if the Young's modulus or mass density of a shaft of a cue is large, the ball trajectory immediately after the impact is deviated largely and then is curved with a large radius of curvature. Because the extended method can evaluate quantitatively this kind of effects of material properties, it is useful for a design of a cue.

Key words: Impact, Cue, Billiards, Curved Ball Trajectory, Efficient Numerical Method, Simulation

1. Introduction

Billiards is one of the most popular games in the world, but little research has been done on the design of a cue, which consists of a leather tap, a plastic ferrule and a wooden shaft. Although it is important how to make a cue easier to use or how to fit a cue to the player's skill, it has been relied on the trial and error method, and a lot of time and cost has been needed. In order to make a reasonable design for a cue, it is desired to develop an effective numerical method and to investigate the effects of material properties of a cue on

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ball motions.

In the previous papers [(1),(2)], we developed an effective numerical method by combining the finite different and the finite element methods to analyze the behaviors of impact between a cue and a ball in the basic follow or draw shot, where the ball is hit on its upper or lower central part in the horizontal or an inclined direction, respectively. Not only the follow and the draw shots but also various ball hitting techniques are required depending on an aspect of a game. In the recent paper [(3)], we extended the numerical method further to treat the spinning case, where the ball is hit on the left or right part in the horizontal direction.

In this paper, the numerical method is extended for the case where the ball is hit on its upper left or right part in an inclined direction to give it a curved trajectory. After the extended method is verified using a high speed camera, a numerical simulation is performed to investigate the effects of stiffness and density of a shaft on the ball motions. The rotation around the vertical axis is referred to as 'spin', and the rotation around the horizontal axis is referred to as 'roll', in this paper.

2. Outline of Numerical Method

We consider the case where a ball is hit on its upper left part with velocity v in an inclined direction φ as shown in Fig.1. We adopt a fixed rectangular coordinate O-x,y,z, where the y axis is taken in the vertical direction and the z axis is taken perpendicular to the cue. The impact force is assumed to be a concentrated point force. At first, the impact force is assumed to be prescribed. The cue is assumed to be isotropically elastic and is analyzed with the finite element method, while the ball and table are assumed to be rigid and the ball is analyzed with the finite difference method as in the previous papers [(1)-(3)]. Then, by coupling the velocities at the impact point, the impact force is modified using the downhill simplex method [(4)].

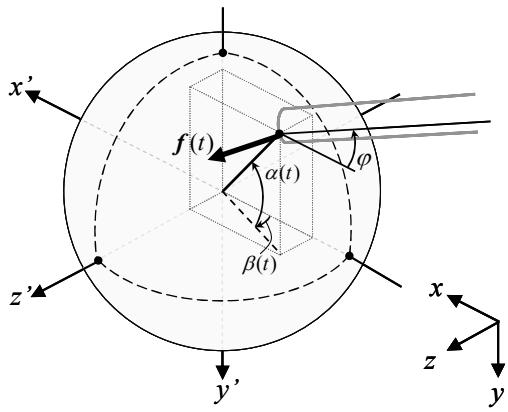


Fig.1 Ball hit on its upper left part in inclined direction

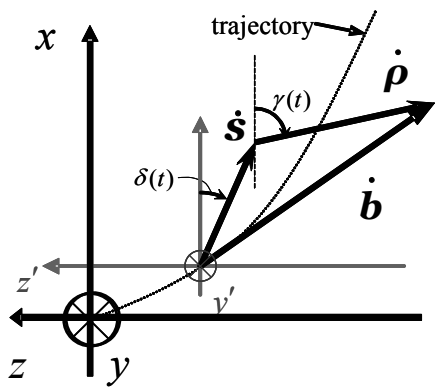


Fig. 2 Schematic view of ball trajectory

3. Finite Difference Analysis of Ball

3.1 Equations of Motion

We adopt not only the fixed rectangular coordinate O-x,y,z, but also

the moving rectangular coordinate $O'-x', y', z'$ which is obtained by translating the O-x,y,z to the centre of a moving ball, as shown in Fig.1. Let $\boldsymbol{b}(t)=(b_x(t), b_y(t), b_z(t))$ denote the displacement of the ball centre, $\boldsymbol{\theta}(t)=(\theta_x(t), \theta_y(t), \theta_z(t))$ the rotational angle of a ball, $f(t)=(f_x(t), f_y(t), f_z(t))$ the impact force and $F_r=(F_{rx}(t), 0, F_{rz}(t))$ the frictional force. The equations of motion are given with Eqs (1)-(6).

For translation:

$$M\ddot{b}_x = f_x(t) - F_{rx}(t) \tag{1}$$

$$M\ddot{b}_y = 0 \tag{2}$$

$$M\ddot{b}_z = f_z(t) - F_{rz}(t) \tag{3}$$

For rotation:

$$\frac{2}{5}MR^2\ddot{\theta}_x = -f_y(t)R\cos\alpha(t)\sin\beta(t) - f_z(t)R\sin\alpha(t) + M_r\sin\gamma(t) - F_{rz}(t)R \tag{4}$$

$$\frac{2}{5}MR^2\ddot{\theta}_y = f_x(t)R\cos\alpha(t)\sin\beta(t) + f_z(t)R\cos\alpha(t)\cos\beta(t) - M_s \tag{5}$$

$$\frac{2}{5}MR^2\ddot{\theta}_z = f_x(t)R\sin\alpha(t) + F_{rx}(t)R - f_y(t)R\cos\alpha(t)\cos\beta(t) - M_r\cos\gamma(t) \tag{6}$$

where M and R are the mass and radius of ball, M_r and M_s are the frictional moments for rolling and spinning, $\alpha(t)$ and $\beta(t)$ are the angles representing the impact point as shown in Fig.1, and ($\dot{}$) represents the time derivative.

The velocity of ball centre, $\dot{\boldsymbol{b}}$, is expressed as the sum of the velocity due to slip between a ball and a table, $\dot{\boldsymbol{s}}$, and the velocity due to rolling, $\dot{\boldsymbol{\rho}}$, as shown in Fig.2, i.e.,

$$\dot{b}_x = \dot{s}_x + R\dot{\theta}_z \tag{7}$$

$$\dot{b}_y = 0 \tag{8}$$

$$\dot{b}_z = \dot{s}_z - R\dot{\theta}_x \tag{9}$$

The $\delta(t)$ and $\gamma(t)$ are the angles between $\dot{\boldsymbol{s}}$ and x' axis, and between $\dot{\boldsymbol{\rho}}$ and x' axis, respectively, i.e.

$$\dot{\rho}\cos\gamma(t) = R\dot{\theta}_z \tag{10}$$

$$\dot{\rho}\sin\gamma(t) = R\dot{\theta}_x \tag{11}$$

$$\dot{s}\cos\delta(t) = \dot{s}_x \tag{12}$$

$$\dot{s}\sin\delta(t) = -\dot{s}_z \tag{13}$$

where $\dot{\rho} = |\dot{\boldsymbol{\rho}}|$ and $\dot{s} = |\dot{\boldsymbol{s}}|$.

The $F_{rx}(t)$ and $F_{rz}(t)$) for $\dot{s} \neq 0$ are represented as

$$F_{rx} = \mu(Mg + f_y(t))\cos\delta(t) \qquad (14)$$

$$F_{rz} = \mu(Mg + f_y(t))\sin\delta(t) \qquad (15)$$

where μ is the coefficient of friction and g is the gravitational acceleration.

3.2 Angles Representing Impact Point

The x', y' and z' coordinates of the impact point are expressed in the terms of $\alpha(t)$ and $\beta(t)$ as follows.

$$x'(t) = -R\cos\alpha(t)\cos\beta(t) \qquad (16)$$

$$y'(t) = -R\sin\alpha(t) \qquad (17)$$

$$z'(t) = R\cos\alpha(t)\sin\beta(t) \qquad (18)$$

Differentiation with t leads to

$$\dot{x}' = R\sin\alpha(t)\cos\beta(t)\dot{\alpha} + R\cos\alpha(t)\sin\beta(t)\dot{\beta} \qquad (19)$$

$$\dot{y}' = -R\cos\alpha(t)\dot{\alpha} \qquad (20)$$

$$\dot{z}' = -R\sin\alpha(t)\sin\beta(t)\dot{\alpha} + R\cos\alpha(t)\cos\beta(t)\dot{\beta} \qquad (21)$$

On the other hand, the $\dot{x}'$, $\dot{y}'$ and $\dot{z}'$ are expressed in terms of $\dot{\theta}_x$, $\dot{\theta}_y$ and $\dot{\theta}_z$ as

$$\dot{x}' = R\cos\alpha(t)\sin\beta(t)\dot{\theta}_y + R\sin\alpha(t)\dot{\theta}_z \qquad (22)$$

$$\dot{y}' = -R\cos\alpha(t)\sin\beta(t)\dot{\theta}_x - R\cos\alpha(t)\cos\beta(t)\dot{\theta}_z \qquad (23)$$

$$\dot{z}' = -R\sin\alpha(t)\dot{\theta}_x + R\cos\alpha(t)\cos\beta(t)\dot{\theta}_y \qquad (24)$$

From Eqs(19)-(21) and Eqs(22)-(24), the following relationship between $\dot{\alpha}$, $\dot{\beta}$ and $\dot{\theta}_x$, $\dot{\theta}_y$, $\dot{\theta}_z$ is obtained.

$$\cos\alpha(t)\sin\beta(t)\dot{\beta} + \sin\alpha(t)\cos\beta(t)\dot{\alpha} = \cos\alpha(t)\sin\beta(t)\dot{\theta}_y + \sin\alpha(t)\dot{\theta}_z \qquad (25)$$

$$\dot{\alpha} = \sin\beta(t)\dot{\theta}_x + \cos\beta(t)\dot{\theta}_z \qquad (26)$$

$$\cos\alpha(t)\cos\beta(t)\dot{\beta} - \sin\alpha(t)\sin\beta(t)\dot{\alpha} = -\sin\alpha(t)\dot{\theta}_x + \cos\alpha(t)\cos\beta(t)\dot{\theta}_y \qquad (27)$$

3.3 Analysis Method

We assume here that the impact force $\boldsymbol{f}(t) = (f_x(t), f_y(t), f_z(t))$ is given. The μ is assumed to be constant for simplicity, and the M_r is considered to be negligibly small because the ball rotates easily even on a slightly inclined table, as in the previous papers [(1)-(3)].

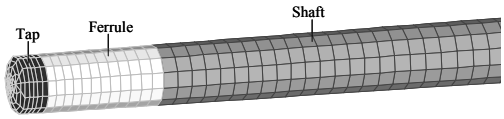
At first, the frictional force $\boldsymbol{F}_r = (F_{rx}(t), 0, F_{rz}(t))$ is obtained from Eqs (14) and (15) by

assuming $\dot{s}(t) \neq 0$. The number of unknowns to be solved is 15, and they are $b_x(t)$,$b_y(t)$,$b_z(t)$, $\theta_x(t)$, $\theta_y(t)$, $\theta_z(t)$, $s_x(t)$, $s_y(t)$,$s_z(t)$, $s(t)$, $\rho(t)$, $\delta(t)$, $\gamma(t)$, $\alpha(t)$ and $\beta(t)$. The $\alpha(t)$ and $\beta(t)$ can be related to $\theta_x(t)$, $\theta_y(t)$ and $\theta_z(t)$ with Eqs(25)-(27), and $b_y(t)= s_y(t)=0$. The $s(t)$ $\rho(t)$, $\delta(t)$ and $\gamma(t)$ are related to $s_x(t)$, $s_z(t)$, $\theta_x(t)$ and $\theta_z(t)$ through Eqs(10)-(13), and $s_x(t)$and $s_z(t)$ can be expressed with $b_x(t)$,$b_z(t)$, $\theta_x(t)$ and $\theta_z(t)$. Thus, the 15 unknowns are reduced to 5 independent unknowns, $b_x(t)$, $b_z(t)$, $\theta_x(t)$, $\theta_y(t)$ and $\theta_z(t)$, which can be solved from the 5 differential equations, Eqs(1), (3)-(6) by the finite difference method (Newmark β=1/6) [5]. The initial conditions are given as the 5 unknowns and their time derivatives are zeros.

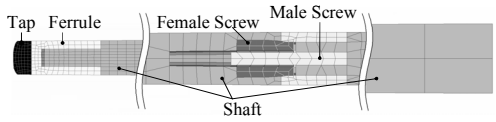
If the first assumption, i.e., $\dot{s}(t) \neq 0$, is found not satisfied, $F_{rx}(t)$ and $F_{rz}(t)$ become unknowns, but $\dot{s}_x(t)$ and $\dot{s}_z(t)$ are known ($\dot{s}_x(t)=\dot{s}_z(t)=0$). Because the number of unknowns (=5) is unchanged, it is possible to solve the equations of motion, Eqs(1), (3)-(6).

4. Finite Element Analysis of Cue

Because the details of the finite element analysis are described in the previous paper[3], only a brief description necessary for the present paper will be given here. A cue is subject to the three dimensional impact force $\boldsymbol{f}(t)=(f_x(t), f_y(t), f_z(t))$, where t is time and the subscripts x, y and z represent the x, y and z components, respectively. The displacement at the impact point is referred to as $\boldsymbol{c}(t)=(c_x(t), c_y(t), c_z(t))$, and was calculated with an elasto-dynamic finite element method (Nastran [6]). We assumed that the cue was supported freely, because the constraint may be not so strong.



(a) Finite element mesh of main part of cue



(b) Cross-section

Fig. 3 Finite element discretization of cue

The cue was discretized with hexagonal or pentagonal elements (total number of elements nodes were 6363 and 6184, respectively). The main part of the finite element mesh is shown in Fig.3. The mesh size near the contact point was chosen to be nearly equal to the radius of the Hertz contact area [7] to reduce the calculation error due to the assumption of a concentrated force instead of a distributed one.

5. Coupling velocities of cue and ball at impact point

We assume that there is no slip at the impact point between a cue and a ball, because the slip is usually avoided by applying chalk on the tip of the tap to increase the frictional force. By equating the velocities of the cue and the ball at the contact point, following

equations are obtained,

$$v\cos\varphi - \dot{c}_x - \left\{\dot{b}_x + R\cos\alpha(t)\sin\beta(t)\dot{\theta}_y + R\sin\alpha(t)\dot{\theta}_z\right\} = 0 \quad (28)$$

$$v\sin\varphi + \dot{c}_y - \left\{-R\cos\alpha(t)\sin\beta(t)\dot{\theta}_x - R\cos\alpha(t)\cos\beta(t)\dot{\theta}_z\right\} = 0 \quad (29)$$

$$-\dot{c}_z - \left\{\dot{b}_z - R\sin\alpha(t)\dot{\theta}_x + R\cos\alpha(t)\cos\beta(t)\dot{\theta}_y\right\} = 0 \quad (30)$$

where the positive directions of $\dot{c}_x$ and $\dot{c}_z$ are taken opposite to those of the x and the z axes in Fig.1. The last two terms in the braces { } in these equations are the x', y' and z' components of the velocity due to rotation at the impact point on the ball as shown with Eqs (22)-(24).

Making use of Eqs (28)-(30), the $f_x(t)$, $f_y(t)$ and $f_z(t)$ are determined as follows. We consider to determine $f_x(t_{n+1})$, $f_y(t_{n+1})$ and $f_z(t_{n+1})$ (n=0,1,2,…, $t_{n+1}= t_n + \Delta t$), when $f_x(t_n)$ $f_y(t_n)$ and $f_z(t_n)$ ($f_x(t_0)=f_y(t_0)=f_z(t_0)=0$) are known.

(a) Assume $f_x(t_{n+1})$, $f_y(t_{n+1})$ and $f_z(t_{n+1})$. Employ the Lagrange method using the values at t_{n-2} ,t_{n-1} and t_{n+1} for $2 \le n$ to reduce the iterative number [8] .

(b) Calculate $c_x(t_{n+1})$, $c_y(t_{n+1})$, $c_z(t_{n+1})$ and their time derivatives with the finite element method (section 4)

(c) Calculate $b_x(t_{n+1})$, $b_z(t_{n+1})$, $\theta_x(t_{n+1})$, $\theta_y(t_{n+1})$, $\theta_z(t_{n+1})$, the other unknowns and their time derivatives with the finite difference method (section 3)

(d) Evaluate the square sum of residuals of the left hand sides of Eqs (28) - (30).

(e) Employ the downhill simplex method[4] with the cost function defined as the above square sum of residuals, and modify $f_x(t_{n+1})$, $f_y(t_{n+1})$ and $f_z(t_{n+1})$.

(f) Repeat (a)-(e) until the cost function becomes small enough.

(g) Repeat above procedures until $\boldsymbol{F}(t_{n+1})$ becomes small enough.

All unknowns including the ball motion during impact are calculated in the above procedures. The ball motion after impact is obtained by solving Eqs (1)-(15) after substituting $\boldsymbol{F}(t)=0$.

6. Verification of Analytical Result

6.1 Experimental Verification

An experiment where a ball was hit on its upper left part in an inclined direction was conducted. The displacement of the ball was measured by taking a movie from above using a high speed camera (Nobby tech Ltd make, Phantom v4.2 model, 2000 frames/s, 0.19 Mega pixels). A still photo was also taken from horizontal direction using a usual digital camera (Nikon D50, 6.10 Mega pixels). The angles $\beta(0)$ and φ were measured directly from the movie and photo, while $\alpha(0)$and v were obtained indirectly from them by making use of the geometrical relationship. The results were $\alpha(0)=12^\circ$, $\beta(0)=20^\circ$, $\varphi=12^\circ$ and v= 1.5 m/s (See Fig.1). The radius of the ball was R=0.0275 m and its mass M=0.169kg.

In the numerical prediction, the coefficient of friction was assumed as μ =0.2, the frictional moments for rolling was assumed as M_r =0, as in the previous papers(1)-(3). The frictional moments for spinning, M_s was taken as M_s=0.022 Nm by performing a preliminary experiment, where a movie was taken using the same high speed camera from above a ball spinning without rolling and sliding, and the decreasing rate of the spinning velocity was measured. The material properties of the cue are the same as those for the analysis 1 ('Normal') in the subsection 7.1, and the time increment in the finite difference method was taken as $\Delta t = 5\times 10^{-5}$s, as in the previous papers(1)-(3).

The experimental and numerical ball trajectories for $0 \le t \le 0.5$ s are compared in Fig.4, where the displacement in the z direction is enlarged 200 times, because they are much small compared with that in the x direction. The symbol representing measurement error (|------|) indicates the length corresponding to 1 pixel. It is found from this result that the ball trajectory predicted numerically agrees well with the

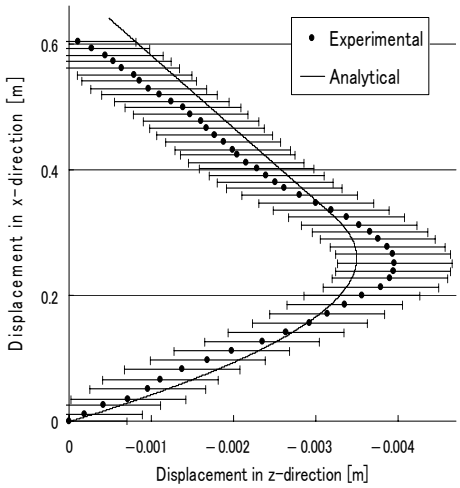


Fig.4 Comparison of analytical of and experimental results of ball trajectories

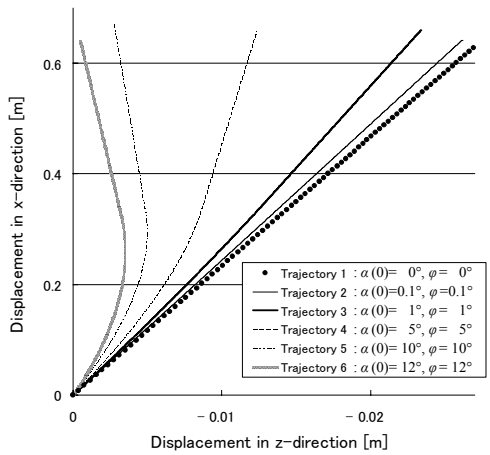


Fig.5 Effect of impact condition on ball trajectories

Table1 Material constants of cue used for analysis

	Parts	Young's Modulus E(GPa)	Poisson's Ratio	Mass Density (kg/m^3)	Wave Velocity in Bar $\sqrt{E/\rho}$ (m/s)
Analysis1 (Normal)	Shaft	12	0.1	602	4464.7
Analysis2 (Soft)	Shaft	6	0.1	301	4464.7
Analysis3 (Stiff)	Shaft	24	0.1	1204	4464.7
Analysis4 (Light)	Shaft	12	0.1	301	6314
Analysis5 (Heavy)	Shaft	12	0.1	1204	3157

one obtained experimentally within the measurement error. It is noted that the ball trajectory is deviated from the x direction immediately after the impact, and then is curved.

6.2 Relationship between Impact Direction and Ball Trajectory

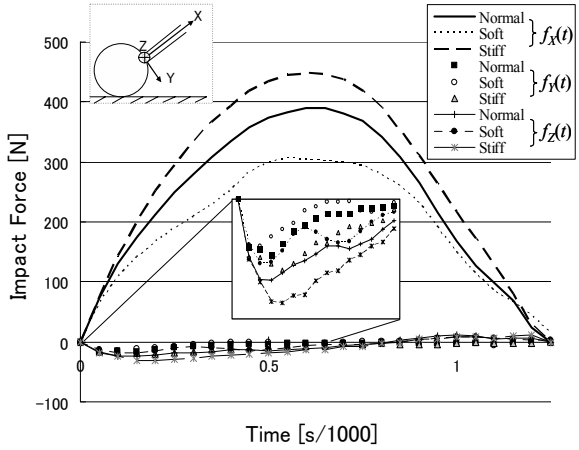
Numerical analyses were carried out using the present method for various values of $\alpha(0)$ and φ, and the ball trajectories obtained are shown in Fig.5, where v and $\beta(0)$ are assumed as v= 1.5 m/s and $\beta(0)$= 20°. It is seen from the figure that the radius of curvature of the ball trajectory is large for the impact on its almost exactly left part with a slightly inclined cue, and this result agrees with an empirical one.

In the previous paper [3], an experiment on the movement of the ball hit at $\alpha(0)$=0, $\beta(0)$= 20° in the direction φ= 0 was performed, and it was observed that the ball trajectory was straight, but deviated from the x direction (See Fig.1). The present numerical result shown in Fig.5 is consistent with the experimental one in the previous paper [3].

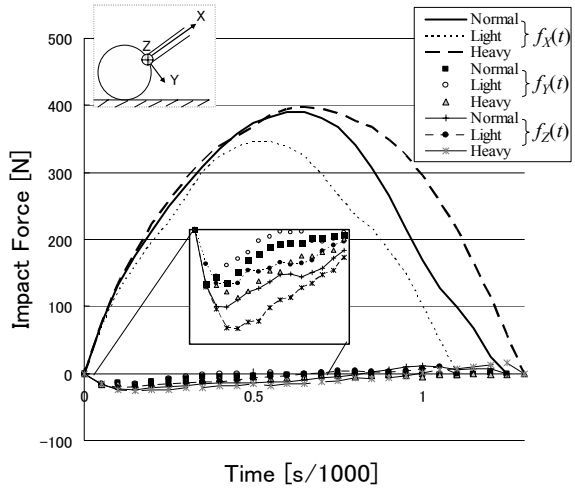
7. Effects of Material Properties of Shaft

7.1 Material Properties

In order to investigate the



(a) Results of analyses 1,2 and 3



(b) Results of analyses 1, 4 and 5

Fig.6 Effect of material properties on impact force

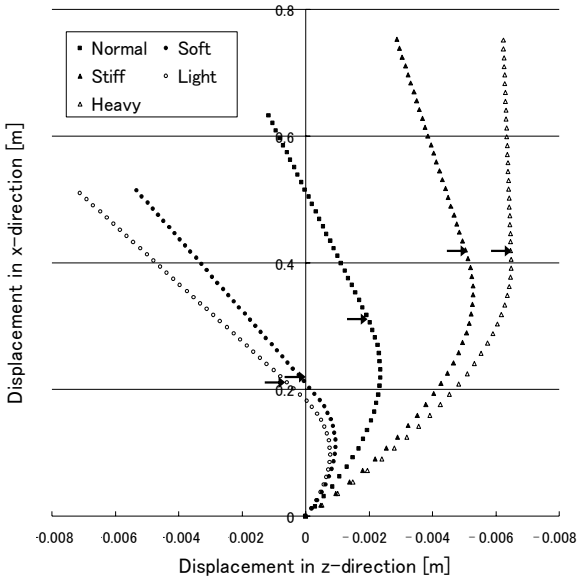
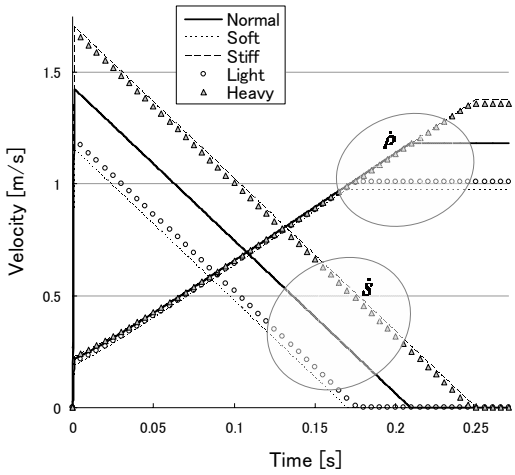


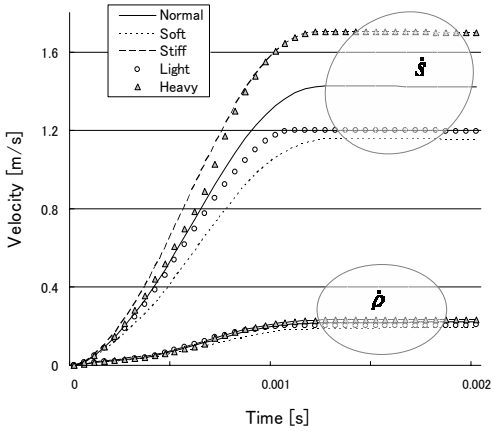
Fig.7 Effect of material properties on ball trajectory

effects of material properties of a shaft of a cue, numerical analyses were performed for $\alpha(0)=15^\circ$, $\beta(0)=20^\circ$, $\varphi=15^\circ$ and $v=1.5$ m/s. Five kinds of material constants of a shaft shown in Table 1 were used. The following material constants of the tap and the ferrule were common to each cue ; Young's modulus E=0.28 GPa, Poison's ratio ν =0.4, mass density=420 kgm^{-3} for the tap, and E=2.57 GPa, ν =0.4, mass density=1074 kgm^{-3} for the ferrule. The other constants were the same as those in subsection 6.1.

In Table 1, both the Young's modulus and the mass density of a shaft were chosen 0.5 and 2 times large for analyses 2 ('Soft') and 3 ('Stiff') respectively, compared with those for analysis 1 ('Normal'). This choice results in the same wave velocity in bar, as shown in the last column. The mass density for analyses 4 ('Light') and 5 ('Heavy') were chosen 0.5 and 2 times larger than analysis 1 ('Normal'), respectively. The wave velocity in bar is different for these analyses as in the last column.



(a) $0 < t < 0.27$ s



(b) $0 < t < 2\times10^{-3}$ s

Fig.8 Effect of material properties on velocities due to sliding and rotation

7.2 Impact Force and Ball Trajectory

The time variations of the impact force are shown in Fig.6(a) for analyses 1-3, and in Fig.6(b) for analyses 1,4,5. It is seen from these figures that the absolute value of longitudinal component of impact force, $|f_X(t)|$, is much larger than the other components for every analysis, it increases with increase in the Young's modulus, and the duration time of impact decreases with increase in the wave velocity in bar.

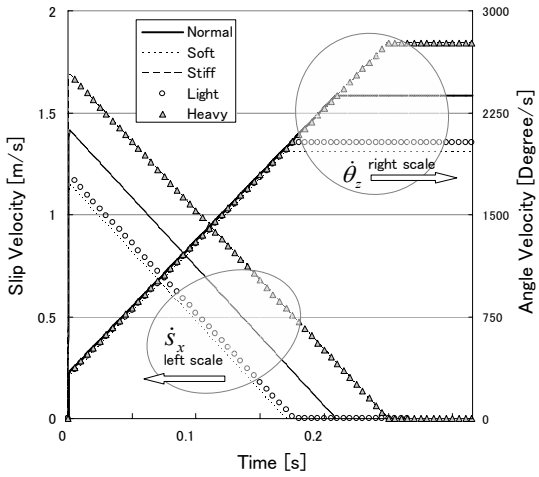
The ball trajectories for $0 \le t \le 0.5$ s are shown in Fig.7, where the displacements in the z direction are enlarged 50 times compared with those in the x direction. It is observed that the ball trajectory is deviated from the x direction immediately after the impact, then it is curved, and it becomes straight after a while. The arrows in the figure indicate the points where the curved trajectories change to the straight ones. It is found that the ball velocity, the deviation of the the ball

trajectory and the radius of curvature of the ball trajectory immediately after the impact increase with increase in the Young's modulus and density of a shaft.

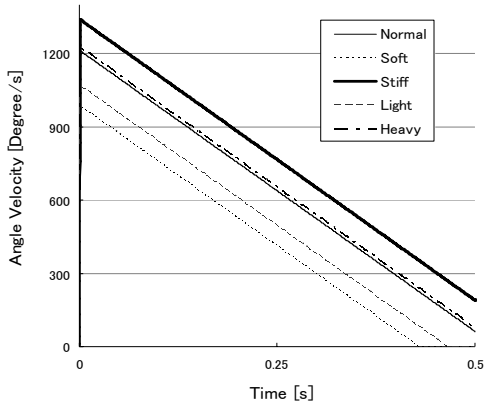
7.3 Slip and Rotational Velocities

The time variations of the velocities of a ball due to the slip between a ball and a table $\dot{s}$ and the rotation $\dot{\rho}$ are shown in Fig.8(a) for $0 \le t \le 0.27$ s and in Fig.8(b) for $0 \le t \le 2\times10^{-3}$ s. It is seen that the $\dot{s}$ increases rapidly within a short impact time (about 10^{-3} s) and then decreases linearly up to zero, and $\dot{s}$ is large for a stiff or heavy shaft. It is also seen that $\dot{\rho}$ increases linearly after a rapid increase in a short impact time and then it becomes constant from the time when $\dot{s}$ becomes zero. This time corresponds to the arrow in Fig.7. The *x, y* and *z* components of $\dot{s}$ and $\dot{\rho}$ are shown in Figs 9 (a), (b) and (c). It is found from Fig.9 (a) that $\dot{\theta}_z$ and $\dot{s}_x$ have similar tendencies to those in Fig.8(a). The spinning angular velocity $\dot{\theta}_y$ is found from Fig.9 (b) to show a similar tendency to that of $\dot{s}$ in Fig.8, although it remains longer than $\dot{s}$.

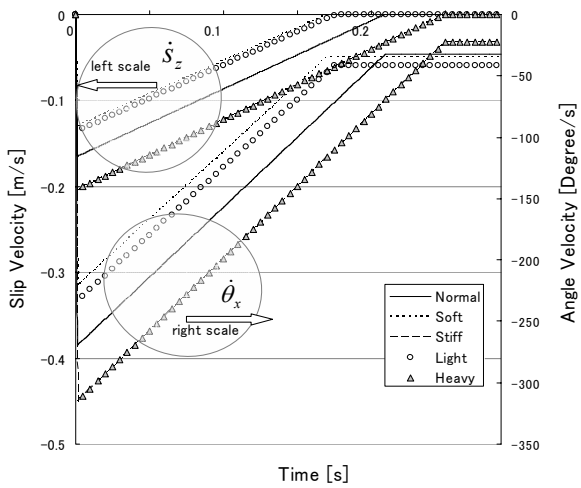
The reason why a ball is deviated in the $-z$ direction immediately after the impact and then returns back in the $+z$ direction after a while can be explained from Fig.9(c) as follows. For example, we consider the case of Analysis 1



(a) $\dot{\theta}_z$ and $\dot{s}_x$



(b) $\dot{\theta}_y$ (spinning angular velocity)



(c) $\dot{\theta}_x$ and $\dot{s}_z$

Fig.9 Effect of material properties on velocity components due to sliding and rotation

('Normal'). Because $\dot{s}_z$ =−0. 16 m/s and $R\dot{\theta}_x$=−0. 06 m/s immediately after the impact (Note $|\dot{s}_z| > |R\dot{\theta}_x|$), the ball is deviated in $-z$ direction at first. After a while, $\dot{s}$ decreases to zero and $\dot{\theta}_x$ becomes a negative constant value. The ball, therefore, returns back in the $+z$ direction.

8. Conclusions

The impact characteristics of a cue for the case where a ball was hit on its upper left (or right) part to make a curved trajectory were numerically analyzed by extending the effective method developed in the previous papers(1)-(3). In the extended method, the dynamic response of a cue was calculated with the elastic finite element method. The ball motion was analyzed with the finite difference method by expressing the ball velocity as the sum of the velocity due to a roll and the velocity due to a slip (between a ball and a table) in a different direction. The finite element analysis for a cue and the finite difference analysis for a ball were coupled by equating the velocities of a cue and a ball at the impact point. The method was verified experimentally by using a high speed camera.

Numerical simulations were carried out using the extended method for several sets of shaft materials and main results are as follows:

1. A ball trajectory becomes curved when the hitting point on a ball and the inclination of a cue are high.
2. The ball velocity, the deviation of the ball trajectory from the hitting direction and the radius of curvature of the ball trajectory immediately after the impact increase with increase in the Young's modulus and density of a shaft.
3. A ball rotates at first with a slip between a ball and a table in a different direction (the trajectory is curved during this period) and then rotates without any slip making a straight trajectory, and the point where the slip disappears is farther for larger Young's modulus and density of a shaft.
4. The impact force increases with increase in the stiffness or the mass density of a shaft, and its duration time decreases with increase in the wave velocity through a shaft.

The extended numerical method can be usefully used for designing a cue, because effects of not only material properties but also geometries of the tap, ferrule and shaft of a cue on the impact force, the ball velocity and the ball trajectory can be quantitatively predicted.

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